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Course: Cybersecurity

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Title: Social Engineering Toolkit (SET)
and Reconnaissance

Exercise 5: Username Enumeration with Sherlock

Objective

The objective of this exercise was to use Sherlock, an Open-Source Intelligence (OSINT) tool, to identify usernames across multiple online platforms. The exercise demonstrates how publicly available information can be collected to understand an individual's digital footprint and highlights the importance of protecting personal information from reconnaissance activities.

Environment

Item	Description
Host Operating System	Arch Linux (Main Boot OS)
Tool Used	Sherlock
Python Version	Python 3
Shell	Bash
Network	Internet Connection
Purpose	Open-Source Intelligence (OSINT) Reconnaissance

Procedure

Step 1: Verify Sherlock Installation

Sherlock was accessed from the terminal, and the help menu was displayed to verify the installation

and review the available options.

Command

```
cd ~/sherlock
```

```
pip install --upgrade pip
```

Observation

The help menu displayed successfully, showing the supported command-line options such as output formatting, site filtering, timeout settings, and export options.

```
(venv) [abrish@archlinux sherlock]$ sherlock --help
usage: sherlock [-h] [--version] [--verbose] [--folderoutput FOLDEROUTPUT] [--output OUTPUT] [--csv] [--xlsx] [--site SITE_NAME]
               [--proxy PROXY_URL] [--dump-response] [--json JSON_FILE] [--timeout TIMEOUT] [--print-all] [--print-found]
               [--no-color] [--browse] [--local] [--nsfw] [--txt] [--ignore-exclusions]
               USERNAMES [USERNAMES ...]

Sherlock: Find Usernames Across Social Networks (Version 0.16.1)

positional arguments:
  USERNAMES              One or more usernames to check with social networks. Check similar usernames using {?} (replace to '_', '-',
                        '.').

options:
  -h, --help            show this help message and exit
  --version             Display version information and dependencies.
  --verbose, -v, -d, --debug
                        Display extra debugging information and metrics.
  --folderoutput, -fo FOLDEROUTPUT
                        If using multiple usernames, the output of the results will be saved to this folder.
  --output, -o OUTPUT   If using single username, the output of the result will be saved to this file.
  --csv                Create Comma-Separated Values (CSV) File.
  --xlsx              Create the standard file for the modern Microsoft Excel spreadsheet (xlsx).
  --site SITE_NAME     Limit analysis to just the listed sites. Add multiple options to specify more than one site.
  --proxy, -p PROXY_URL
                        Make requests over a proxy. e.g. socks5://127.0.0.1:1080
  --dump-response      Dump the HTTP response to stdout for targeted debugging.
  --json, -j JSON_FILE
                        Load data from a JSON file or an online, valid, JSON file. Upstream PR numbers also accepted.
  --timeout TIMEOUT    Time (in seconds) to wait for response to requests (Default: 60)
  --print-all         Output sites where the username was not found.
  --print-found        Output sites where the username was found (also if exported as file).
  --no-color           Don't color terminal output
  --browse, -b         Browse to all results on default browser.
  --local, -l          Force the use of the local data.json file.
  --nsfw              Include checking of NSFW sites from default list.
  --txt               Enable creation of a txt file
  --ignore-exclusions
                        Ignore upstream exclusions (may return more false positives)
(venv) [abrish@archlinux sherlock]$ _
```

Figure 1: Sherlock Help Menu.

Step 2: Perform a Single Username Search

A search was performed for a single username across all supported platforms.

Command

```
python3 sherlock johndoe123
```

Observation

Sherlock searched hundreds of supported websites and displayed whether the username existed on each platform. Existing accounts returned positive matches, while unavailable accounts were reported as not found.

```
(venv) └─# sherlock johndoe123
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username johndoe123 on:

[+] 9GAG: https://www.9gag.com/u/johndoe123
[+] About.me: https://about.me/johndoe123
[+] Airbit: https://airbit.com/johndoe123
[+] AllMyLinks: https://allmylinks.com/johndoe123
[+] Anilist: https://anilist.co/user/johndoe123/
[+] Archive of Our Own: https://archiveofourown.org/users/johndoe123
[+] ArtStation: https://www.artstation.com/johndoe123
[+] Audiojungle: https://audiojungle.net/user/johndoe123
[+] Behance: https://www.behance.net/johndoe123
[+] Bluesky: https://bsky.app/profile/johndoe123.bsky.social
[+] BoardGameGeek: https://boardgamegeek.com/user/johndoe123
[+] BraveCommunity: https://community.brave.com/u/johndoe123/
[+] BugCrowd: https://bugcrowd.com/johndoe123
[+] BuyMeACoffee: https://buymeacoff.ee/johndoe123
[+] CGTrader: https://www.cgtrader.com/johndoe123
[+] Carrd: https://johndoe123.carrd.co/
[+] Career.habr: https://career.habr.com/johndoe123
[+] Chess: https://www.chess.com/member/johndoe123
[+] Clapper: https://clapperapp.com/johndoe123
[+] Clozemaster: https://www.clozemaster.com/players/johndoe123
[+] Clubhouse: https://www.clubhouse.com/@johndoe123
[+] Codecademy: https://www.codecademy.com/profiles/johndoe123
[+] Codechef: https://www.codechef.com/users/johndoe123
[+] Codeforces: https://codeforces.com/profile/johndoe123
[+] Codepen: https://codepen.io/johndoe123
[+] Codewars: https://www.codewars.com/users/johndoe123
[+] ColourLovers: https://www.colourlovers.com/lover/johndoe123
[+] Cplusplus: https://cplusplus.com/user/johndoe123
[+] Cracked Forum: https://cracked.ax/johndoe123
```

```
[+] Vero: https://vero.co/johndoe123
[+] VirusTotal: https://www.virustotal.com/gui/user/johndoe123
[+] Warrior Forum: https://www.warriorforum.com/members/johndoe123.html
[+] Wattpad: https://www.wattpad.com/user/johndoe123
[+] Wikidot: http://www.wikidot.com/user:info/johndoe123
[+] Wikipedia: https://en.wikipedia.org/wiki/Special:CentralAuth/johndoe123?uselang=qqx
[+] WordPressOrg: https://profiles.wordpress.org/johndoe123/
[+] Wordnik: https://www.wordnik.com/users/johndoe123
[+] Wykop: https://www.wykop.pl/ludzie/johndoe123
[+] Xbox Gamertag: https://xboxgamertag.com/search/johndoe123
[+] YouTube: https://www.youtube.com/@johndoe123
[+] akniga: https://akniga.org/profile/johndoe123
[+] couchsurfing: https://www.couchsurfing.com/people/johndoe123
[+] fixya: https://www.fixya.com/users/johndoe123
[+] furaffinity: https://www.furaffinity.net/user/johndoe123
[+] geocaching: https://www.geocaching.com/p/default.aspx?u=johndoe123
[+] jbz.com.pl: https://jbzd.com.pl/uzytkownik/johndoe123
[+] jeuxvideo: https://www.jeuxvideo.com/profil/johndoe123
[+] kofi: https://ko-fi.com/johndoe123
[+] kwork: https://kwork.ru/user/johndoe123
[+] mastodon.social: https://mastodon.social/@johndoe123
[+] minds: https://www.minds.com/johndoe123/
[+] nairaland.com: https://www.nairaland.com/johndoe123
[+] osu!: https://osu.ppy.sh/users/johndoe123
[+] pikabu: https://pikabu.ru/@johndoe123
[+] Pinterest: https://www.pinterest.com/johndoe123/
[+] pr0gramm: https://pr0gramm.com/user/johndoe123
[+] tumblr: https://johndoe123.tumblr.com/
[+] Platzi: https://platzi.com/p/johndoe123/
[+] BabyRu: https://www.baby.ru/u/johndoe123
[+] Wowhead: https://wowhead.com/user=johndoe123

[*] Search completed with 150 results
(venv) └─
```

Figure 2: Single Username Search Results.

Step 3: Perform a Multiple Username Search

Multiple usernames were searched simultaneously.

Command

```
python3 sherlock johndoe123 janedoe456 bobsmith789
```

Observation

Sherlock processed each username sequentially and displayed the results separately for every target.

```
(venv) └─# sherlock johndoe123 janedoe456 bobsmith789
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username johndoe123 on:

[+] 9GAG: https://www.9gag.com/u/johndoe123
[+] About.me: https://about.me/johndoe123
[+] Airbit: https://airbit.com/johndoe123
[+] AllMyLinks: https://allmylinks.com/johndoe123
[+] Anilist: https://anilist.co/user/johndoe123/
[+] Archive of Our Own: https://archiveofourown.org/users/johndoe123
[+] ArtStation: https://www.artstation.com/johndoe123
[+] Audiojungle: https://audiojungle.net/user/johndoe123
[+] Behance: https://www.behance.net/johndoe123
[+] Bluesky: https://bsky.app/profile/johndoe123.bsky.social
[+] BoardGameGeek: https://boardgamegeek.com/user/johndoe123
[+] BraveCommunity: https://community.brave.com/u/johndoe123/
[+] BugCrowd: https://bugcrowd.com/johndoe123
[+] BuyMeACoffee: https://buymeacoff.ee/johndoe123
[+] CGTrader: https://www.cgtrader.com/johndoe123
[+] Carrd: https://johndoe123.carrd.co/
[+] Career.habr: https://career.habr.com/johndoe123
[+] Chess: https://www.chess.com/member/johndoe123
[+] Clapper: https://clapperapp.com/johndoe123
[+] Clozemaster: https://www.clozemaster.com/players/johndoe123
[+] Clubhouse: https://www.clubhouse.com/@johndoe123
[+] Codecademy: https://www.codecademy.com/profiles/johndoe123
[+] Codechef: https://www.codechef.com/users/johndoe123
[+] Codeforces: https://codeforces.com/profile/johndoe123
[+] Codepen: https://codepen.io/johndoe123
[+] Codewars: https://www.codewars.com/users/johndoe123
[+] ColourLovers: https://www.colourlovers.com/lover/johndoe123
[+] Cplusplus: https://cplusplus.com/user/johndoe123
[+] Cracked Forum: https://cracked.ax/johndoe123
```

```

[*] Checking username janedoe456 on:

[+] Chess: https://www.chess.com/member/janedoe456
[+] Codecademy: https://www.codecademy.com/profiles/janedoe456
[+] Discord: https://discord.com
[+] Disqus: https://disqus.com/janedoe456
[+] Duolingo: https://www.duolingo.com/profile/janedoe456
[+] Genius (Users): https://genius.com/janedoe456
[+] HudsonRock: https://cavalier.hudsonrock.com/api/json/v2/osint-tools/search-by-username?usern
[+] Kick: https://kick.com/janedoe456
[+] SoundCloud: https://soundcloud.com/janedoe456
[+] Star Citizen: https://robertsspaceindustries.com/citizens/janedoe456
[+] TikTok: https://www.tiktok.com/@janedoe456
[+] Trakt: https://www.trakt.tv/users/janedoe456
[+] Trello: https://trello.com/janedoe456
[+] Velomania: https://forum.velomania.ru/member.php?username=janedoe456
[+] Venmo: https://account.venmo.com/u/janedoe456
[+] Xbox Gamertag: https://xboxgamertag.com/search/janedoe456
[+] YouTube: https://www.youtube.com/@janedoe456
[+] kofi: https://ko-fi.com/janedoe456
[+] last.fm: https://last.fm/user/janedoe456
[+] tumblr: https://janedoe456.tumblr.com/
[+] BabyRu: https://www.baby.ru/u/janedoe456

[*] Checking username bobsmith789 on:

[+] GitLab: https://gitlab.com/bobsmith789
[+] Gravatar: http://en.gravatar.com/bobsmith789
[+] Imgur: https://imgur.com/user/bobsmith789
[+] Medium: https://medium.com/@bobsmith789
[+] Memrise: https://www.memrise.com/user/bobsmith789/
[+] Periscope: https://www.periscope.tv/bobsmith789/
[+] Smule: https://www.smule.com/bobsmith789
[+] Snapchat: https://www.snapchat.com/add/bobsmith789
[+] TikTok: https://www.tiktok.com/@bobsmith789
[+] Telegram: https://t.me/bobsmith789
[+] Trakt: https://www.trakt.tv/users/bobsmith789
[+] Velomania: https://forum.velomania.ru/member.php?username=bobsmith789
[+] Wikipedia: https://en.wikipedia.org/wiki/Special:CentralAuth/bobsmith789?uselang=qqx
[+] Xbox Gamertag: https://xboxgamertag.com/search/bobsmith789

```

```

[+] Trakt: https://www.trakt.tv/users/janedoe456
[+] Trello: https://trello.com/janedoe456
[+] Velomania: https://forum.velomania.ru/member.php?username=janedoe456
[+] Venmo: https://account.venmo.com/u/janedoe456
[+] Xbox Gamertag: https://xboxgamertag.com/search/janedoe456
[+] YouTube: https://www.youtube.com/@janedoe456
[+] kofi: https://ko-fi.com/janedoe456
[+] last.fm: https://last.fm/user/janedoe456
[+] tumblr: https://janedoe456.tumblr.com/
[+] BabyRu: https://www.baby.ru/u/janedoe456

[*] Checking username bobsmith789 on:

[+] GitLab: https://gitlab.com/bobsmith789
[+] Gravatar: http://en.gravatar.com/bobsmith789
[+] Imgur: https://imgur.com/user/bobsmith789
[+] Medium: https://medium.com/@bobsmith789
[+] Memrise: https://www.memrise.com/user/bobsmith789/
[+] Periscope: https://www.periscope.tv/bobsmith789/
[+] Smule: https://www.smule.com/bobsmith789
[+] Snapchat: https://www.snapchat.com/add/bobsmith789
[+] TikTok: https://www.tiktok.com/@bobsmith789
[+] Telegram: https://t.me/bobsmith789
[+] Trakt: https://www.trakt.tv/users/bobsmith789
[+] Velomania: https://forum.velomania.ru/member.php?username=bobsmith789
[+] Wikipedia: https://en.wikipedia.org/wiki/Special:CentralAuth/bobsmith789?uselang=qqx
[+] Xbox Gamertag: https://xboxgamertag.com/search/bobsmith789
[+] YouTube: https://www.youtube.com/@bobsmith789
[+] Pinterest: https://www.pinterest.com/bobsmith789/
[+] BabyRu: https://www.baby.ru/u/bobsmith789

[*] Search completed with 189 results
(env) └─#
(env) └─# _

```

Figure 3: Multiple Username Search.

Step 4: Export Search Results

The search results were exported to a folder for documentation and future analysis.

Command

```
python3 sherlock johndoe123 --folderoutput ~/sherlock-results
```

Observation

Sherlock created the output directory and saved the results as text files containing the discovered platforms and associated profile URLs.

```
(venv) └─$ sherlock johndoe123 --folderoutput ~/sherlock-results
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username johndoe123 on:

[+] 9GAG: https://www.9gag.com/u/johndoe123
[+] About.me: https://about.me/johndoe123
[+] Academia.edu: https://independent.academia.edu/johndoe123
[+] Airbit: https://airbit.com/johndoe123
[+] AllMyLinks: https://allmylinks.com/johndoe123
[+] Anilist: https://anilist.co/user/johndoe123/
[+] Archive of Our Own: https://archiveofourown.org/users/johndoe123
[+] ArtStation: https://www.artstation.com/johndoe123
[+] Audiojungle: https://audiojungle.net/user/johndoe123
[+] Behance: https://www.behance.net/johndoe123
[+] Bluesky: https://bsky.app/profile/johndoe123.bsky.social
[+] BoardGameGeek: https://boardgamegeek.com/user/johndoe123
[+] BraveCommunity: https://community.brave.com/u/johndoe123/
[+] BugCrowd: https://bugcrowd.com/johndoe123
[+] BuyMeACoffee: https://buymeacoff.ee/johndoe123
[+] CGTrader: https://www.cgtrader.com/johndoe123
[+] Carrd: https://johndoe123.carrd.co/
[+] Career.habr: https://career.habr.com/johndoe123
[+] Chess: https://www.chess.com/member/johndoe123
[+] Clozemaster: https://www.clozemaster.com/players/johndoe123
[+] Clubhouse: https://www.clubhouse.com/@johndoe123
[+] Codecademy: https://www.codecademy.com/profiles/johndoe123
[+] Codechef: https://www.codechef.com/users/johndoe123
[+] Codeforces: https://codeforces.com/profile/johndoe123
[+] Codewars: https://www.codewars.com/users/johndoe123
[+] ColourLovers: https://www.colourlovers.com/lover/johndoe123
[+] Cplusplus: https://cplusplus.com/user/johndoe123
[+] Cracked Forum: https://cracked.ax/johndoe123
[+] Crowdin: https://crowdin.com/profile/johndoe123
```

```
[+] Vero: https://vero.co/johndoe123
[+] VirusTotal: https://www.virustotal.com/gui/user/johndoe123
[+] Warrior Forum: https://www.warriorforum.com/members/johndoe123
[+] Wattpad: https://www.wattpad.com/user/johndoe123
[+] Wikidot: http://www.wikidot.com/user:info/johndoe123
[+] Wikipedia: https://en.wikipedia.org/wiki/Special:Centralauth/redirect/johndoe123
[+] WordPressOrg: https://profiles.wordpress.org/johndoe123/
[+] Wordnik: https://www.wordnik.com/users/johndoe123
[+] Wykop: https://www.wykop.pl/ludzie/johndoe123
[+] Xbox Gamertag: https://xboxgamertag.com/search/johndoe123
[+] YouTube: https://www.youtube.com/@johndoe123
[+] akniga: https://akniga.org/profile/johndoe123
[+] couchsurfing: https://www.couchsurfing.com/people/johndoe123
[+] fixya: https://www.fixya.com/users/johndoe123
[+] furaffinity: https://www.furaffinity.net/user/johndoe123
[+] geocaching: https://www.geocaching.com/p/default.aspx?u=johndoe123
[+] jbz.com.pl: https://jbz.com.pl/uzytkownik/johndoe123
[+] jeuxvideo: https://www.jeuxvideo.com/profil/johndoe123
[+] kofi: https://ko-fi.com/johndoe123
[+] kwork: https://kwork.ru/user/johndoe123
[+] mastodon.social: https://mastodon.social/@johndoe123
[+] minds: https://www.minds.com/johndoe123/
[+] nairaland.com: https://www.nairaland.com/johndoe123
[+] osu!: https://osu.ppy.sh/users/johndoe123
[+] pikabu: https://pikabu.ru/@johndoe123
[+] Pinterest: https://www.pinterest.com/johndoe123/
[+] pr0gramm: https://pr0gramm.com/user/johndoe123
[+] tumblr: https://johndoe123.tumblr.com/
[+] Platzi: https://platzi.com/p/johndoe123/
[+] BabyRu: https://www.baby.ru/u/johndoe123
[+] Wowhead: https://wowhead.com/user=johndoe123

[*] Search completed with 148 results
(venv) └─# _
```

```
(venv) └─# ls -l ~/sherlock-results
total 8
-rw-r--r-- 1 abrish abrish 5694 Jul 27 05:59 johndoe123.txt
(venv) └─# cat ~/sherlock-results/johndoe123.txt
https://www.9gag.com/u/johndoe123
https://about.me/johndoe123
https://airbit.com/johndoe123
https://allmylinks.com/johndoe123
https://anilist.co/user/johndoe123/
https://archiveofourown.org/users/johndoe123
https://www.artstation.com/johndoe123
https://audiojungle.net/user/johndoe123
https://www.behance.net/johndoe123
https://bsky.app/profile/johndoe123.bsky.social
https://boardgamegeek.com/user/johndoe123
https://community.brave.com/u/johndoe123/
https://bugcrowd.com/johndoe123
https://buymeacoff.ee/johndoe123
https://www.cgtrader.com/johndoe123
https://johndoe123.carrd.co/
https://career.habr.com/johndoe123
https://www.chess.com/member/johndoe123
https://clapperapp.com/johndoe123
https://www.clozemaster.com/players/johndoe123
https://www.clubhouse.com/@johndoe123
https://www.codecademy.com/profiles/johndoe123
https://www.codechef.com/users/johndoe123
https://codeforces.com/profile/johndoe123
https://www.codewars.com/users/johndoe123
https://www.colourlovers.com/lover/johndoe123
https://cplusplus.com/user/johndoe123
https://cracked.ax/johndoe123
https://crowdin.com/profile/johndoe123
https://dev.to/johndoe123
```

Figure 4: Exported Sherlock Results.

Step 5: Perform a Focused Search

The search scope was limited to selected social media platforms.

Command

```
python3 sherlock johndoe123 --site Instagram --site Twitter --site LinkedIn
```

Observation

Sherlock searched only the specified websites, reducing execution time and focusing the investigation on selected platforms.

```
(venv) └─# sherlock johndoe123 --print-found
A problem occurred while checking for an update: 'tag_name'
[*] Checking username johndoe123 on:

[+] 9GAG: https://www.9gag.com/u/johndoe123
[+] About.me: https://about.me/johndoe123
[+] Airbit: https://airbit.com/johndoe123
[+] AllMyLinks: https://allmylinks.com/johndoe123
[+] Anilist: https://anilist.co/user/johndoe123/
[+] Archive of Our Own: https://archiveofourown.org/users/johndoe123
[+] ArtStation: https://www.artstation.com/johndoe123
[+] AudioJungle: https://audiojungle.net/user/johndoe123
[+] Behance: https://www.behance.net/johndoe123
[+] Bluesky: https://bsky.app/profile/johndoe123.bsky.social
[+] BoardGameGeek: https://boardgamegeek.com/user/johndoe123
[+] BraveCommunity: https://community.brave.com/u/johndoe123/
[+] BugCrowd: https://bugcrowd.com/johndoe123
[+] BuyMeACoffee: https://buymeacoff.ee/johndoe123
[+] CGTrader: https://www.cgtrader.com/johndoe123
[+] Carrd: https://johndoe123.carrd.co/
[+] Career.habr: https://career.habr.com/johndoe123
[+] Chess: https://www.chess.com/member/johndoe123
[+] Clozemaster: https://www.clozemaster.com/players/johndoe123
[+] Clubhouse: https://www.clubhouse.com/@johndoe123
[+] Codecademy: https://www.codecademy.com/profiles/johndoe123
[+] Codechef: https://www.codechef.com/users/johndoe123
[+] Codeforces: https://codeforces.com/profile/johndoe123
[+] Codewars: https://www.codewars.com/users/johndoe123
[+] ColourLovers: https://www.colourlovers.com/lover/johndoe123
[+] Cplusplus: https://cplusplus.com/user/johndoe123
[+] Cracked Forum: https://cracked.ax/johndoe123
[+] CrowdIn: https://crowdin.com/profile/johndoe123
[+] DEV Community: https://dev.to/johndoe123
[+] DailyMotion: https://www.dailymotion.com/johndoe123
```

```

[*] TheMovieDB: https://www.themoviedb.org/u/johndoe123
[*] TikTok: https://www.tiktok.com/@johndoe123
[*] Topmate: https://topmate.io/johndoe123
[*] Telegram: https://t.me/johndoe123
[*] Tellonym.me: https://tellonym.me/johndoe123
[*] Tenor: https://tenor.com/users/johndoe123
[*] TradingView: https://www.tradingview.com/u/johndoe123/
[*] Trakt: https://www.trakt.tv/users/johndoe123
[*] Trello: https://trello.com/johndoe123
[*] Twitch: https://www.twitch.tv/johndoe123
[*] Typeracer: https://data.typeracer.com/pit/profile?user=johndoe123
[*] Ultimate-Guitar: https://ultimate-guitar.com/u/johndoe123
[*] VK: https://vk.com/johndoe123
[*] Velomania: https://forum.velomania.ru/member.php?username=johndoe123
[*] Venmo: https://account.venmo.com/u/johndoe123
[*] Vero: https://vero.co/johndoe123
[*] VirusTotal: https://www.virustotal.com/gui/user/johndoe123
[*] Warrior Forum: https://www.warriorforum.com/members/johndoe123.html
[*] Wattpad: https://www.wattpad.com/user/johndoe123
[*] Wikidot: http://www.wikidot.com/user/info/johndoe123
[*] Wikipedia: https://en.wikipedia.org/wiki/Special:CentralAuth/johndoe123?uselang=qqx
[*] WordPressOrg: https://profiles.wordpress.org/johndoe123/
[*] Wordnik: https://www.wordnik.com/users/johndoe123
[*] Wykop: https://www.wykop.pl/ludzie/johndoe123
[*] Xbox Gamertag: https://xboxgamertag.com/search/johndoe123
[*] YouTube: https://www.youtube.com/@johndoe123
[*] akniga: https://akniga.org/profile/johndoe123
[*] couchsurfing: https://www.couchsurfing.com/people/johndoe123
[*] fixya: https://www.fixya.com/users/johndoe123
[*] furaffinity: https://www.furaffinity.net/user/johndoe123
[*] geocaching: https://www.geocaching.com/p/default.aspx?u=johndoe123
[*] jbsd.com.pl: https://jbsd.com.pl/uzytkownik/johndoe123
[*] jeuxvideo: https://www.jeuxvideo.com/profil/johndoe123
[*] kofi: https://ko-fi.com/johndoe123
[*] kwork: https://kwork.ru/user/johndoe123
[*] mastodon.social: https://mastodon.social/@johndoe123
[*] minds: https://www.minds.com/johndoe123/
[*] nairaland.com: https://www.nairaland.com/johndoe123
[*] osu!: https://osu.ppy.sh/users/johndoe123
[*] pikabu: https://pikabu.ru/@johndoe123
[*] Pinterest: https://www.pinterest.com/johndoe123/
[*] program: https://pr0gramm.com/user/johndoe123
[*] tumblr: https://johndoe123.tumblr.com/
[*] Platzi: https://platzi.com/p/johndoe123/
[*] BabyRu: https://www.baby.ru/u/johndoe123
[*] Wowhead: https://wowhead.com/user=johndoe123

```

```

[*] Search completed with 147 results

```

```

(venv) └─

```

```

(venv) └─# sherlock johndoe123 --site GitHub
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username johndoe123 on:

[+] GitHub: https://www.github.com/johndoe123

[*] Search completed with 1 results
(venv) └─# sherlock johndoe123 --site GitHub --site Reddit --site Instagram
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username johndoe123 on:

[+] GitHub: https://www.github.com/johndoe123
[+] Reddit: https://www.reddit.com/user/johndoe123

[*] Search completed with 2 results
(venv) └─# _

```

Figure 5: Focused Search on Selected Platforms.

Step 6: Reconnaissance Scenario

A fictional target named **Alex Johnson** was used to demonstrate username enumeration.

The following username variations were examined:

```
python3 sherlock alexjohnson alex.johnson ajohnson alexj --folderoutput ~/sherlock-
results
```

Observation

Multiple username variations were analyzed to identify publicly available profiles across different platforms. This demonstrated how individuals may use different username formats on different websites.

```
(venv) [abrish@archlinux sherlock]$ sherlock alexjohnson alex.johnson ajohnson alexj --folderoutput ~/sherlock-results
Update available! 0.16.1 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
Warning: Could not load exclusions, continuing without them.
[*] Checking username alexjohnson on:

[+] 7Cups: https://www.7cups.com/@alexjohnson
[+] 9GAG: https://www.9gag.com/u/alexjohnson
[+] Airliners: https://www.airliners.net/user/alexjohnson/profile/photos
[+] AllMyLinks: https://allmylinks.com/alexjohnson
[+] Apple Developer: https://developer.apple.com/forums/profile/alexjohnson
[+] Apple Discussions: https://discussions.apple.com/profile/alexjohnson
[+] Aparat: https://www.aparat.com/alexjohnson/
[+] Archive.org: https://archive.org/details/@alexjohnson
[+] ArtStation: https://www.artstation.com/alexjohnson
[+] AudioJungle: https://audiojungle.net/user/alexjohnson
[+] Bandcamp: https://www.bandcamp.com/alexjohnson
[+] Behance: https://www.behance.net/alexjohnson
[+] Bluesky: https://bsky.app/profile/alexjohnson.bsky.social
[+] BugCrowd: https://bugcrowd.com/alexjohnson
[+] CGTrader: https://www.cgtrader.com/alexjohnson
[+] Carbonmade: https://alexjohnson.carbonmade.com
[+] Carrd: https://alexjohnson.carrd.co/
[+] Chatujme.cz: https://profil.chatujme.cz/alexjohnson
[+] Chess: https://www.chess.com/member/alexjohnson
[+] Clubhouse: https://www.clubhouse.com/@alexjohnson
[+] Codeberg: https://codeberg.org/alexjohnson
[+] Codecademy: https://www.codecademy.com/profiles/alexjohnson
[+] Codechef: https://www.codechef.com/users/alexjohnson
[+] Codeforces: https://codeforces.com/profile/alexjohnson
[+] Codewars: https://www.codewars.com/users/alexjohnson
[+] Codolio: https://codolio.com/profile/alexjohnson
[+] ColourLovers: https://www.colourlovers.com/lover/alexjohnson
[+] Coroflot: https://www.coroflot.com/alexjohnson
[+] Cracked Forum: https://cracked.ax/alexjohnson
[+] Credly: https://www.credly.com/users/alexjohnson
[+] Crevado: https://alexjohnson.crevado.com
[+] DEV Community: https://dev.to/alexjohnson
[+] DeviantArt: https://www.deviantart.com/alexjohnson
[+] Discord.bio: https://discords.com/api-v2/bio/details/alexjohnson
```

```
[*] Checking username alexj on:

[+] 7Cups: https://www.7cups.com/@alexj
[+] 9GAG: https://www.9gag.com/u/alexj
[+] About.me: https://about.me/alexj
[+] Airlines: https://www.airliners.net/user/alexj/profile/photos
[+] AllMyLinks: https://allmylinks.com/alexj
[+] AniWorld: https://aniworld.to/user/profil/alexj
[+] Anilist: https://anilist.co/user/alexj/
[+] Apple Developer: https://developer.apple.com/forums/profile/alexj
[+] Apple Discussions: https://discussions.apple.com/profile/alexj
[+] Aparat: https://www.aparat.com/alexj/
[+] Archive of Our Own: https://archiveofourown.org/users/alexj
[+] Archive.org: https://archive.org/details/@alexj
[+] ArtStation: https://www.artstation.com/alexj
[+] Ask Fedora: https://ask.fedoraproject.org/u/alexj
[+] Atcoder: https://atcoder.jp/users/alexj
[+] Vjudge: https://VJudge.net/user/alexj
[+] Audiojungle: https://audiojungle.net/user/alexj
[+] Bandcamp: https://www.bandcamp.com/alexj
[+] Behance: https://www.behance.net/alexj
[+] Blogger: https://alexj.blogspot.com
[+] Bookcrossing: https://www.bookcrossing.com/mybookshelf/alexj/
[+] BoardGameGeek: https://boardgamegeek.com/user/alexj
[+] BraveCommunity: https://community.brave.com/u/alexj/
[+] BuyMeACoffee: https://buymeacoff.ee/alexj
[+] CGTrader: https://www.cgtrader.com/alexj
[+] Car Talk Community: https://community.cartalk.com/u/alexj/summary
[+] Carbonmade: https://alexj.carbonmade.com
[+] Carrd: https://alexj.carrd.co/
[+] Championat: https://www.championat.com/user/alexj
[+] Chatujme.cz: https://profil.chatujme.cz/alexj
[+] Chess: https://www.chess.com/member/alexj
[+] Codeberg: https://codeberg.org/alexj
[+] Codecademy: https://www.codecademy.com/profiles/alexj
[+] Codechef: https://www.codechef.com/users/alexj
[+] Codeforces: https://codeforces.com/profile/alexj
[+] Coderwall: https://coderwall.com/alexj
[+] Codewars: https://www.codewars.com/users/alexj
[+] Codolio: https://codolio.com/profile/alexj
```

```
[+] geocaching: https://www.geocaching.com/p/default.aspx?u=alexj
[+] habr: https://habr.com/ru/users/alexj
[+] hackster: https://www.hackster.io/alexj
[+] igromania: http://forum.igromania.ru/member.php?username=alexj
[+] interpals: https://www.interpals.net/alexj
[+] jeuxvideo: https://www.jeuxvideo.com/profil/alexj
[+] kofi: https://ko-fi.com/alexj
[+] kwork: https://kwork.ru/user/alexj
[+] last.fm: https://last.fm/user/alexj
[+] leasehackr: https://forum.leasehackr.com/u/alexj/summary/
[+] livelib: https://www.livelib.ru/reader/alexj
[+] mastodon.cloud: https://mastodon.cloud/@alexj
[+] mastodon.social: https://mastodon.social/@alexj
[+] mstdn.social: https://mstdn.social/@alexj
[+] mercadolibre: https://www.mercadolibre.com.br/perfil/alexj
[+] minds: https://www.minds.com/alexj/
[+] nairaland.com: https://www.nairaland.com/alexj
[+] nnRU: https://alexj.www.nn.ru/
[+] opennet: https://www.opennet.ru/~alexj
[+] osu!: https://osu.ppy.sh/users/alexj
[+] phpRU: https://php.ru/forum/members/?username=alexj
[+] pikabu: https://pikabu.ru/@alexj
[+] Pinterest: https://www.pinterest.com/alexj/
[+] svidbook: https://www.svidbook.ru/user/alexj
[+] threads: https://www.threads.net/@alexj
[+] tumblr: https://alexj.tumblr.com/
[+] Platzi: https://platzi.com/p/alexj/
[+] BabyRu: https://www.baby.ru/u/alexj
[+] Wowhead: https://wowhead.com/user=alexj

[*] Search completed with 648 results
(venv) [abrish@archlinux sherlock]$ ls ~/sherlock-results
johndoe123.txt
```

Figure 6: Reconnaissance Results for Username Variations.

Discussion Points:

- How much information can be gathered from username alone?
- What privacy settings help protect against this reconnaissance?
- How do attackers use this information for targeted attacks?

Discussion Questions

1. How much information can be gathered from a username alone?

A username can reveal a significant amount of publicly available information if it is reused across multiple websites. Investigators may discover social media accounts, developer profiles, personal interests, locations, employment details, and online activity patterns. This information can be combined to build a detailed profile of an individual.

2. What privacy settings help protect against this reconnaissance?

Users can reduce OSINT exposure by making profiles private, limiting profile visibility to trusted contacts, using different usernames for different platforms, disabling search engine indexing where available, and preventing account discovery through phone numbers or email addresses.

3. How do attackers use this information for targeted attacks?

Attackers use publicly available information to conduct spear-phishing, credential stuffing, and social engineering attacks. By understanding a target's interests, job role, and online activity, they can create more convincing phishing emails and impersonation attempts, increasing the likelihood of a successful attack.

Analysis

This exercise demonstrated how Sherlock automates username enumeration across hundreds of websites by checking for publicly available usernames. The tool helps investigators identify an individual's online presence by correlating usernames across multiple platforms.

The exercise also illustrated that consistent username usage across different services can make it easier to build a digital profile using only publicly available information. Security professionals can use this type of reconnaissance to assess an organization's exposure and recommend appropriate privacy practices.

Results

The username enumeration exercise was completed successfully.

The following outcomes were observed:

- Sherlock launched successfully.
- Single username searches completed successfully.
- Multiple usernames were processed efficiently.

- Search results were exported to a designated output directory.
- Focused searches reduced execution time by limiting the number of queried platforms.
- Publicly available usernames were identified across multiple online services.
- The exercise demonstrated how OSINT tools can assist in digital footprint analysis.

Defensive Countermeasures

Organizations and individuals should implement the following measures to reduce OSINT exposure:

- Avoid reusing the same username across multiple online platforms.
- Configure privacy settings to limit publicly visible profile information.
- Review social media accounts regularly to remove unnecessary personal information.
- Conduct periodic OSINT assessments to understand publicly available information.
- Educate employees about the risks associated with oversharing information online.
- Use strong, unique passwords and enable Multi-Factor Authentication (MFA) to protect online accounts.

Conclusion

This exercise demonstrated how Sherlock can be used to perform username enumeration as part of an Open-Source Intelligence (OSINT) investigation. By identifying usernames across multiple online platforms, investigators can better understand an individual's digital footprint and publicly available information. The exercise emphasized the importance of privacy, responsible information sharing, and regular OSINT assessments to reduce exposure and strengthen organizational security.